

Project Title :**Blue Carbon Guardian: AI-Based Coastal Ecosystem Monitoring and Climate Risk Assessment for India****Abstract**

Climate change is causing rising sea levels, coastal erosion, stronger cyclones, and the degradation of India's coastal ecosystems. This project proposes an AI-based web application to monitor and analyze Blue Carbon ecosystems, including mangroves, seagrass, and salt marshes, across India's coastline. The system will collect environmental and climate-related data such as mangrove cover, sea-level rise, coastal erosion, cyclone history, pollution indicators, and other relevant parameters based on data availability. A suitable AI/ML model will be selected to assess the climate risk level of coastal regions. The application is expected to provide an interactive coastal map, carbon estimation, plantation tracking, environmental reporting, and awareness features to support sustainable coastal ecosystem management.

Expected Inputs

- Coastal region/state information
- Mangrove, seagrass, and salt marsh data
- Sea-level rise and coastal erosion data
- Climate parameters (temperature, rainfall, humidity, etc.)
- Cyclone and pollution information
- Plantation activity records
- User-reported environmental issues (optional)

Expected Outputs

- AI-based climate risk assessment
- Interactive map of India's coastal ecosystems
- Carbon storage/estimation insights
- Plantation tracking dashboard
- Environmental issue reports
- Coastal conservation recommendations

Proposed Responsibilities

Member 1 – SAUMYA (24BYB1130)

- Literature survey and dataset collection
- Data preprocessing and exploratory analysis
- AI/ML model selection, training, and evaluation
- Frontend development, dashboard, and map visualization

Member 2 – KENGUA ASINI (24BYB1132)

- Backend and database development
- Plantation tracking and environmental reporting modules
- Carbon estimation module
- AI model integration
- Testing and documentation

Note

The proposed features, AI/ML model, datasets, and division of work are tentative and may be modified during the project development based on data availability and project requirements.